XRPL Validator — Daily Engineering Report · 2026-10-24

Period: 2026-10-24 00:00 to 24:00 US Eastern (2026-10-24 04:00 to 2026-10-25 04:00 UTC); ledgers #107,863,469 to #107,885,426. Every count from the ledger and the validator's logs stops at #107,885,426, closed 03:59:51 UTC. Generated 2026-10-25 00:20 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Normal — coverage outside planned maintenance 100.000% (objective 99.9%), every validation agreed with the network, no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,883 of the 21,958 ledgers the network closed (99.658%); every validation matched the network's validated ledger. Unsigned: 75 during planned restarts.
- **Native engine:** 0 engine receipts.
- **Changes:** 1 deployment (cycle 188), 1 commit merged into main.
- **Independence program:** 0 of 30 qualifying days (snapshot at 04:20:56 UTC).
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-11-04 14:46 UTC.

Service levels

Measure	Value
Ledgers closed by the network	21,958
Validations issued	21,883
Coverage (validations / ledgers closed)	99.658%
Validations matching the network's validated ledger hash	21,883 of 21,883
Unsigned during planned restarts	75
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%
Restarts	1
Basis: our own ledger hash	21,883
External agreement, data.xrpl.org (UTC day 2026-10-24)	0.99645 (75 missed of 21,107)

Restarts:

- 21:18:36 UTC: deployment of cycle 188; validations resumed at #107,879,386 (75 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,863,469–#107,885,426), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters, snapshot at 04:20:56 UTC (since the last restart, 2026-10-24 21:18 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	6,361 ledgers, 1,426,321 transactions	6,361 fully matched; results 0, bytes 0, keys 0/0
2 · transaction records	6,364 ledgers, 1,455,642 records	1,455,642 exact; differ 0, tree 0
3 · transaction order	6,364 ledgers (mode: on)	6,364 exact; order 0, results 0, bytes 0, refused 0
4 · ledger data from peers	6,428 ledgers fetched	6,428 matched; differ 0, failed 0
4 · trusted validations	6,436 ledgers judged	6,436 agreed; differ 0, short of quorum 0
5 · our own ledger hash	6,439 ledgers	6,364 own; differ 0, partial 75
Writer	every ledger written	mismatch 0, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

No receipts this day.

Incidents

No incidents.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
188	`74e298a`	21:19:25 UTC	2,183 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	400 matched, 0 mismatched	#63 from 21:45 UTC: in progress

Cycle 188 delivered: `74e298a` build: note that the SDK checkout's xrpl-core must sign with k256 0.14

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
20:27 UTC	xrpl-validator	`74e298a`	build: note that the SDK checkout's xrpl-core must sign with k256 0.14

Findings fixed

No engine findings merged into main this day.

Docs and operations

- **build: note that the SDK checkout's xrpl-core must sign with k256 0.14** (`74e298a` , xrpl-validator)

Independence program

Qualifying days: **0 of 30** (snapshot at 04:20:56 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-10-24 — tracked #107,859,766–#107,881,783	Value
Validations	21,919 (21,919 own, 0 borrowed)
Coverage of the tracked ledgers	99.55%
Our own transaction order	on
Ledger data from peers	no

Not qualifying: the ledger data came from the reference node, not peers (Step 4).

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	21,958 (#107,863,469–#107,885,426)
Mean close interval	3.93 s
Longest close interval	10 s (before #107,881,843)
Closes at least 15 s apart	0
Empty ledgers	0

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-21 14:46 UTC	2026-11-04 14:46 UTC

Amendments enabled on mainnet: 97.

Open items

- Soak #63 (cycle 188) in progress since 2026-10-24 21:45 UTC; due to complete 2026-10-25 21:45 UTC.
- Amendment fixCleanup3_4_0: activation due ~2026-11-04 14:46 UTC.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
10-24 20:27	10-24 20:27	xrpl-validator	`74e298a`	build: note that the SDK checkout's xrpl-core must sign with k256 0.14

Times in UTC. A commit appears on the day it was committed to main or merged into it.